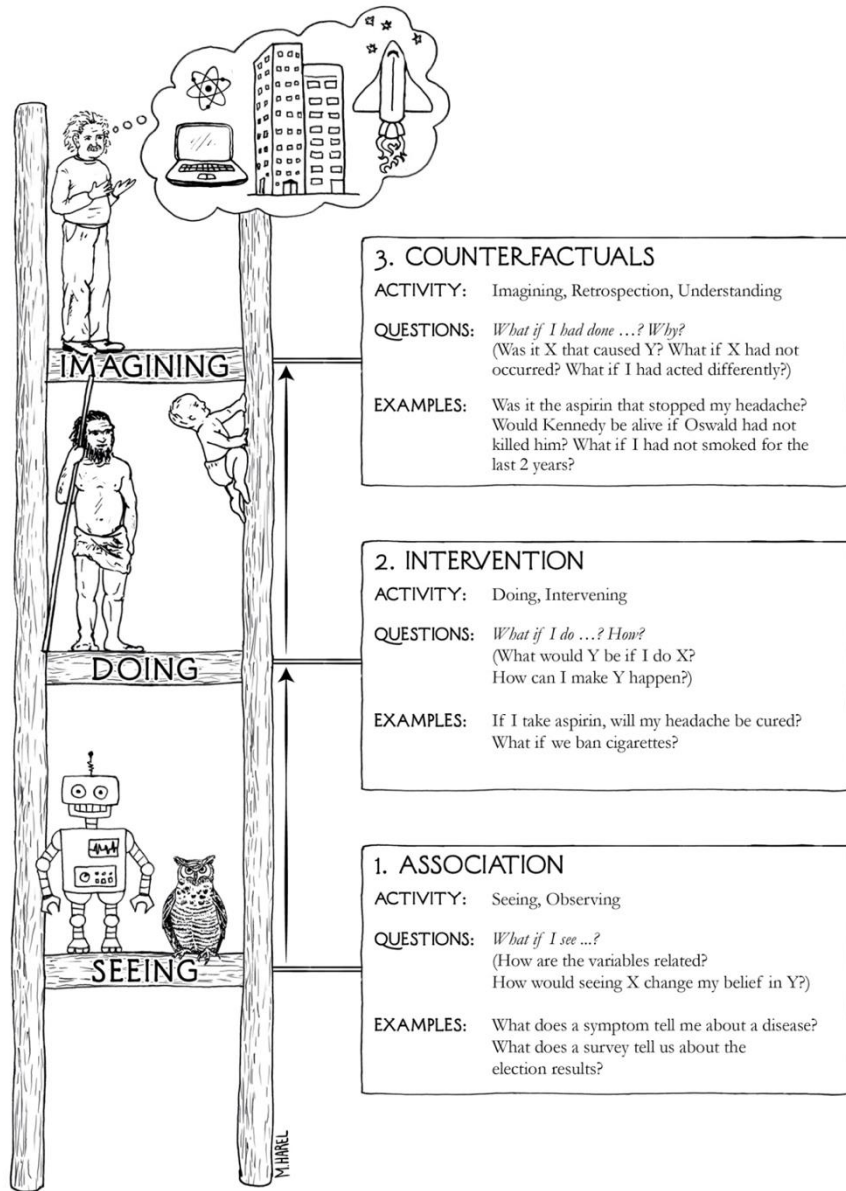


Counterfactuals for **Explainability** and **Fairness** in Machine Learning

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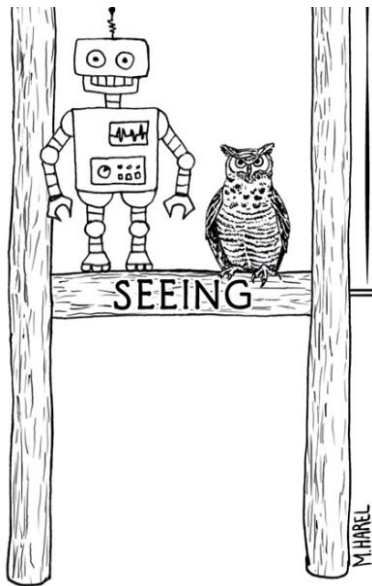


JUDEA PEARL
WINNER OF THE TURING AWARD
AND DANA MACKENZIE

THE BOOK OF WHY



THE NEW SCIENCE
OF CAUSE AND EFFECT

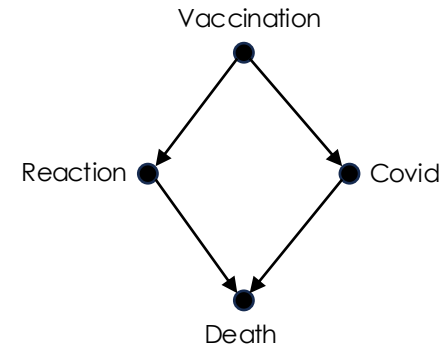


1. ASSOCIATION

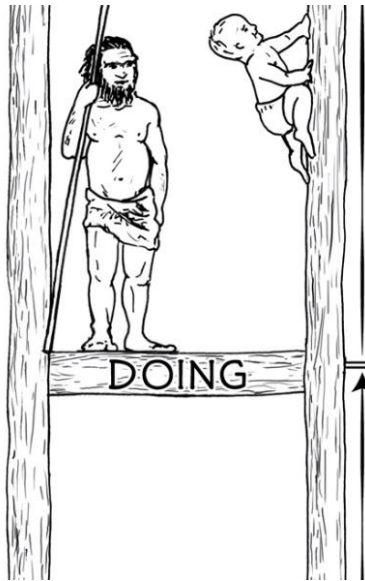
ACTIVITY: Seeing, Observing

QUESTIONS: *What if I see ...?*
(How are the variables related?)
How would seeing X change my belief in Y?)

EXAMPLES: What does a symptom tell me about a disease?
What does a survey tell us about the election results?



- What is the chance that I get a Reaction after Vaccination?
- What is the chance that I Die if I get Vaccinated?

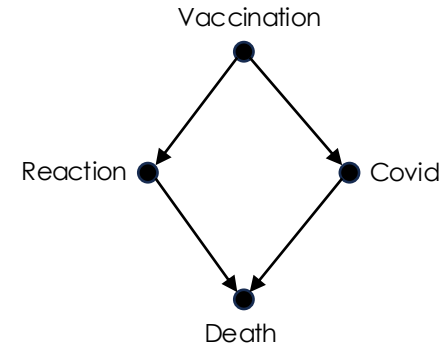


2. INTERVENTION

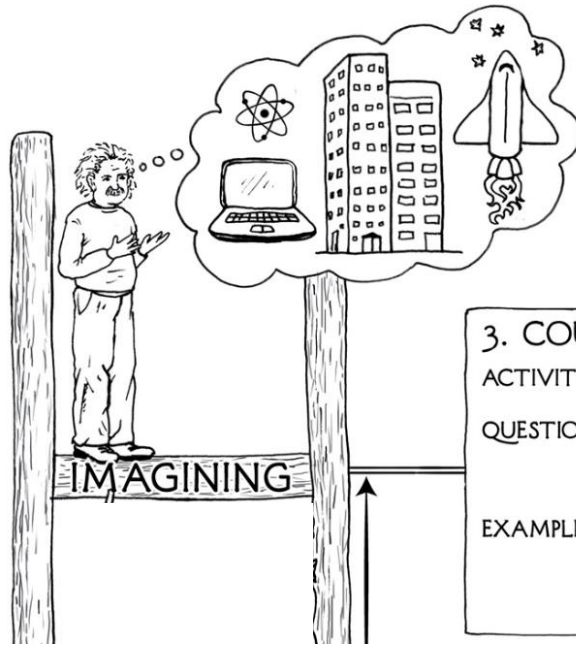
ACTIVITY: Doing, Intervening

QUESTIONS: *What if I do ...? How?*
(What would Y be if I do X?
How can I make Y happen?)

EXAMPLES: If I take aspirin, will my headache be cured?
What if we ban cigarettes?



- What is the efficacy of the Vaccine?
- How less likely I am to Die if I get Vaccinated?

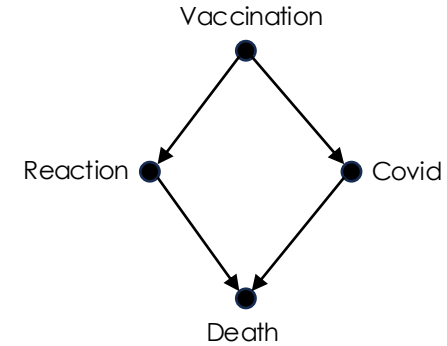


3. COUNTERFACTUALS

ACTIVITY: Imagining, Retrospection, Understanding

QUESTIONS: *What if I had done ...? Why?*
(Was it X that caused Y? What if X had not occurred? What if I had acted differently?)

EXAMPLES: Was it the aspirin that stopped my headache?
Would Kennedy be alive if Oswald had not killed him? What if I had not smoked for the last 2 years?



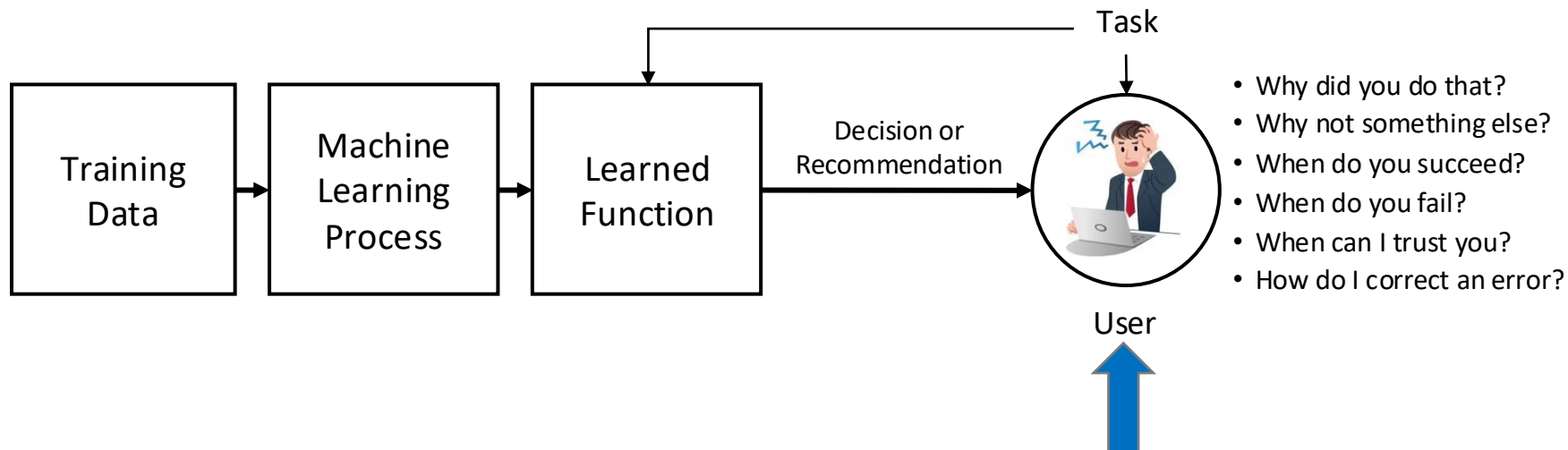
- This person didn't Vaccinate and Died.
Would he have lived if he Vaccinated?

FACTUAL No Vaccination, No Reaction, Covid -> Death

COUNTERFACTUAL Vaccination, Reaction?, Covid? -> Death?

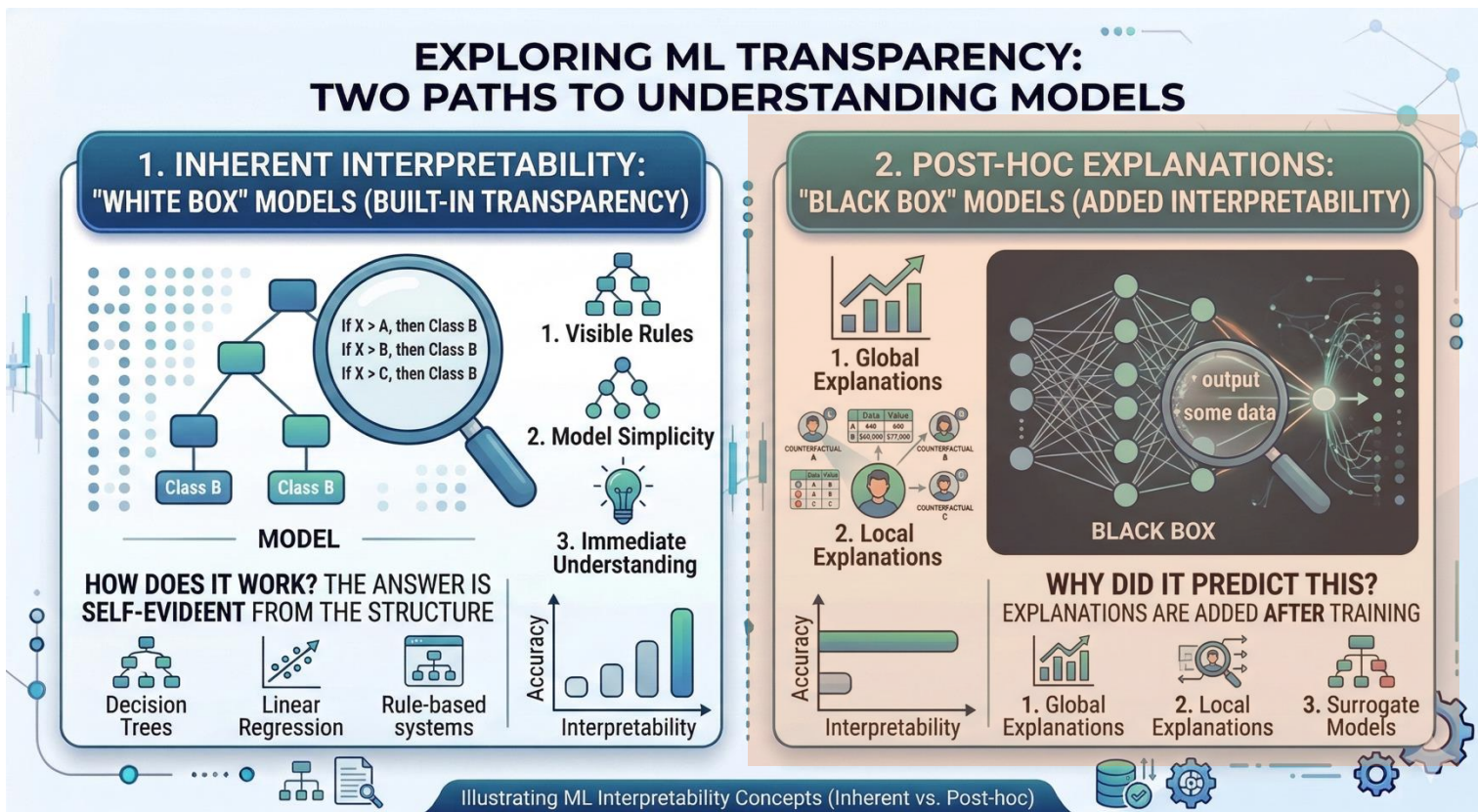
Explainability

Explainable AI (XAI)

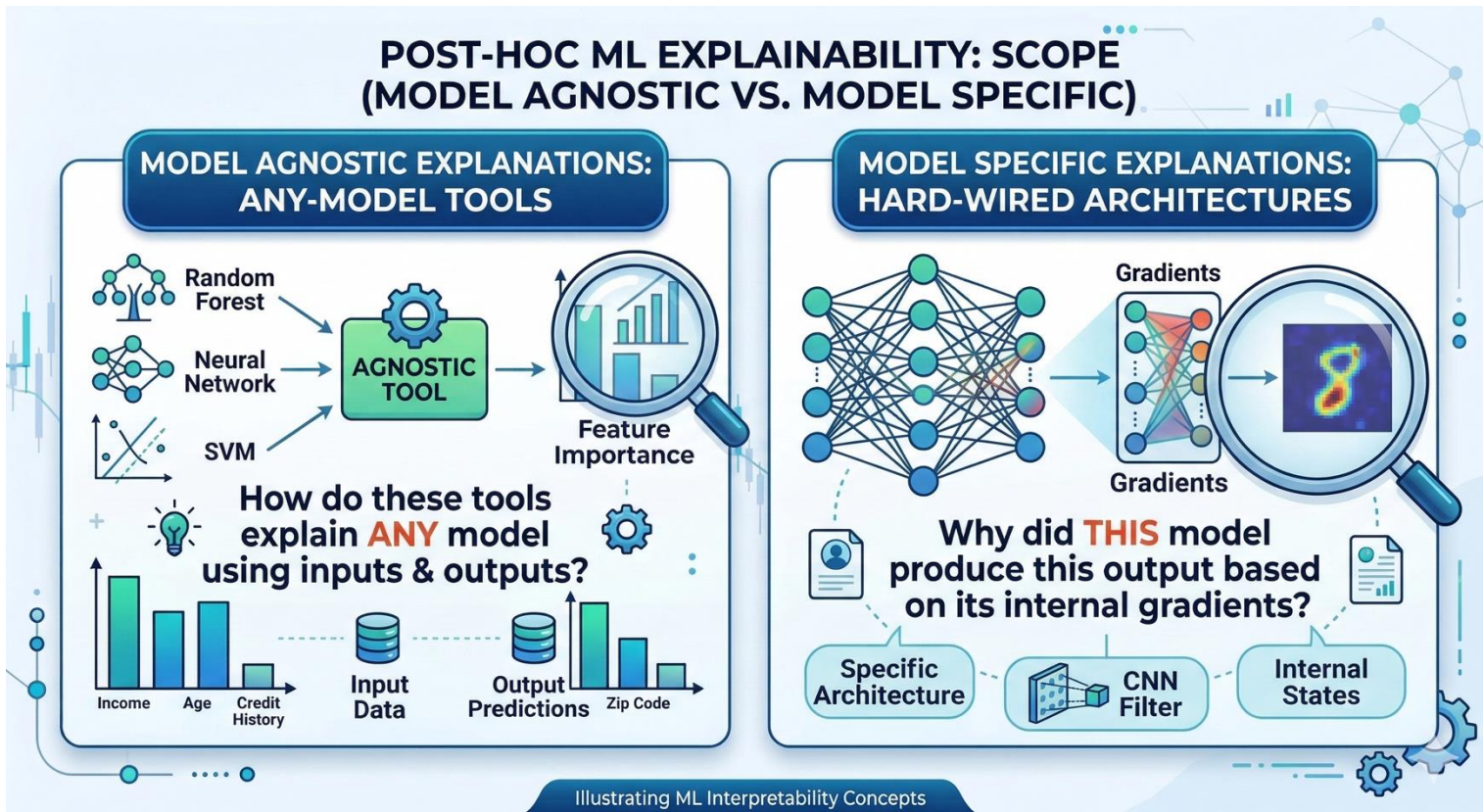


- The target of XAI is an end user who:
 - depends on decisions, recommendations, or actions of the system
 - needs to understand the rationale for the system's decisions to understand, appropriately trust, and effectively manage the system

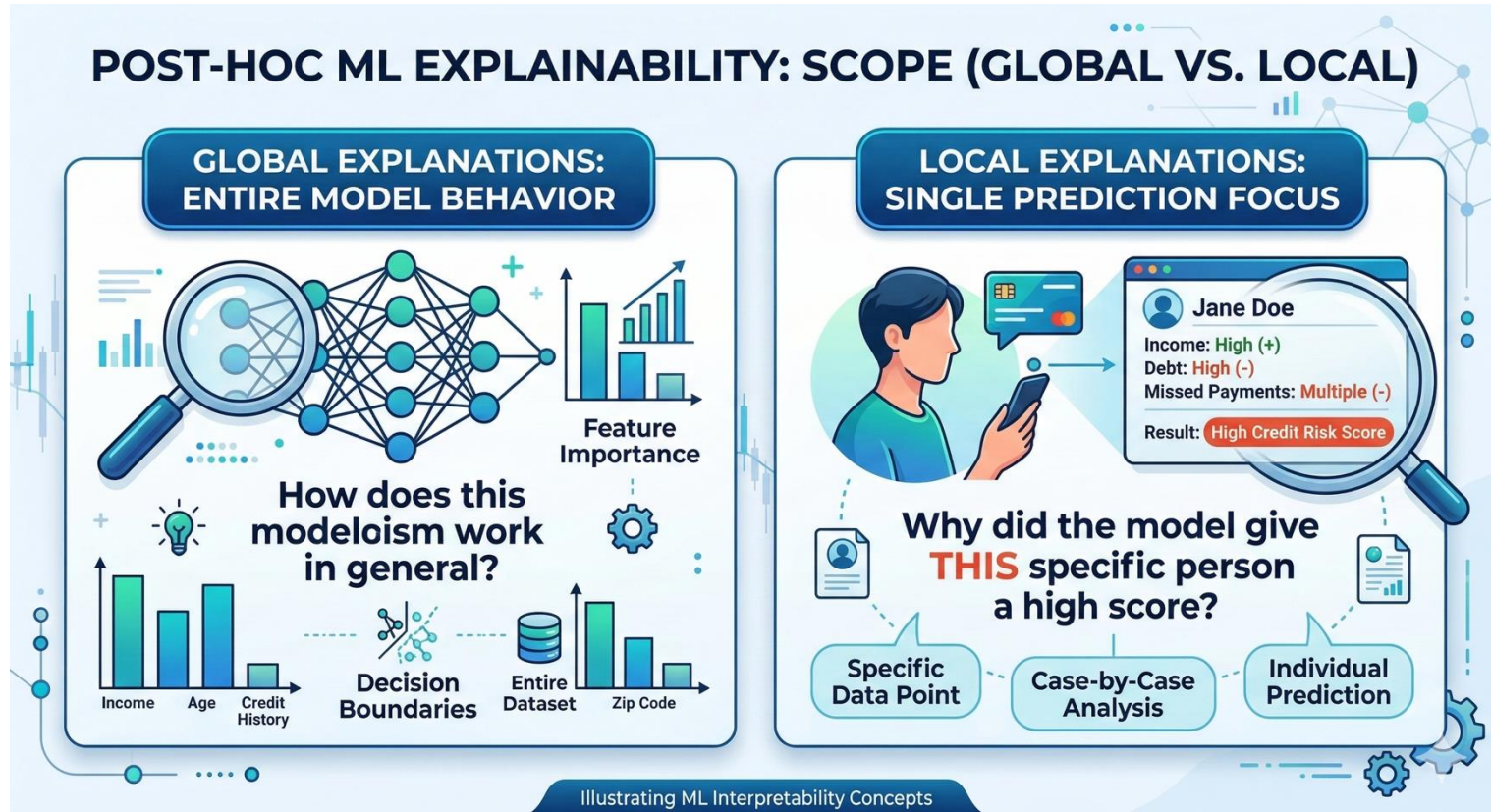
Interpretability vs. Explainability



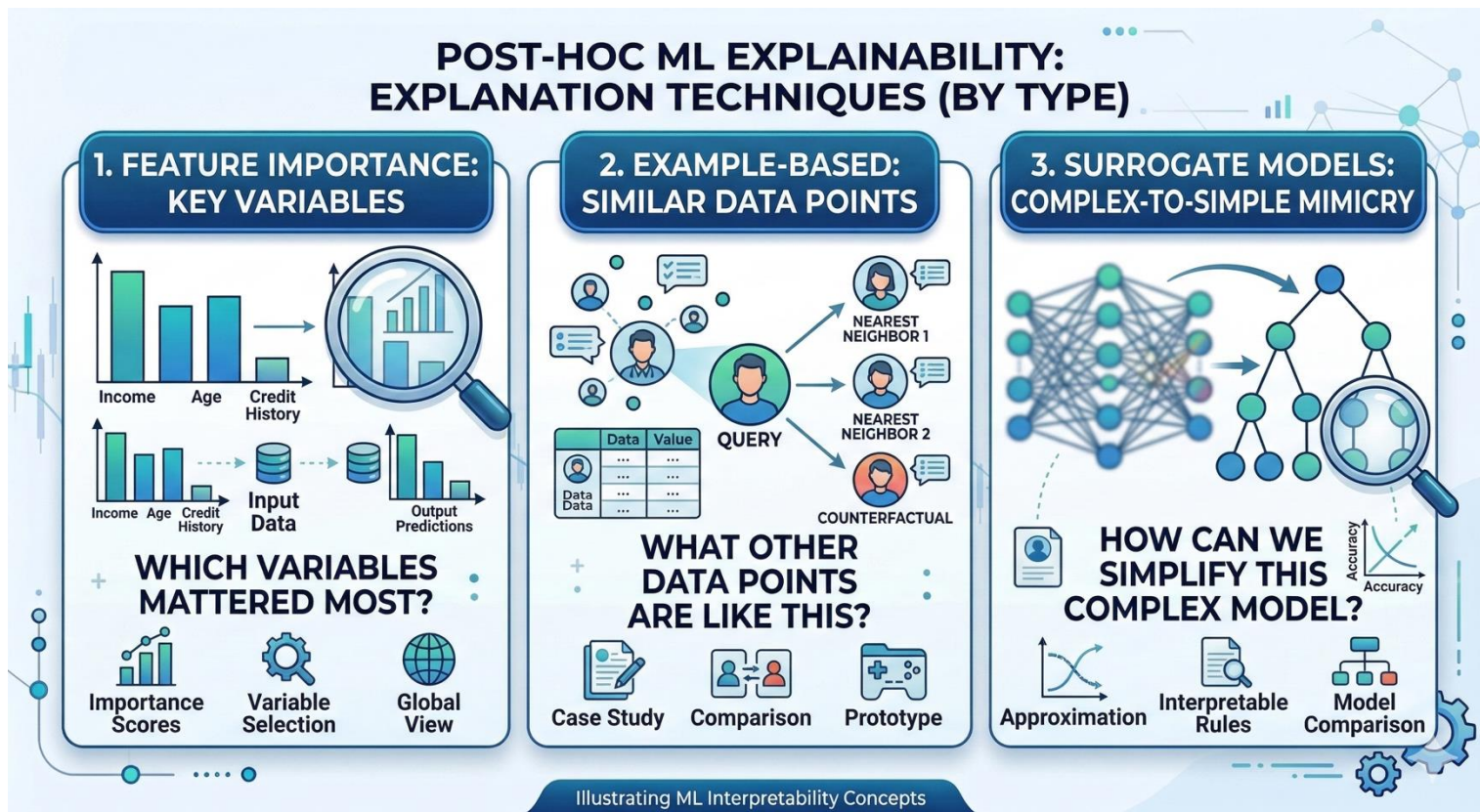
Model Agnostic vs. Model Specific



Global vs. Local



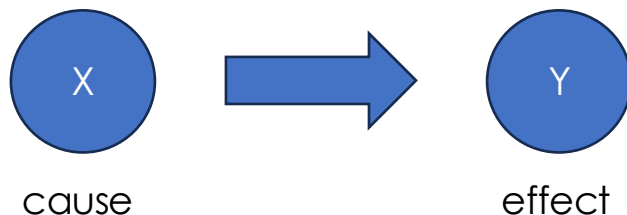
Explanation Type



Explainability

Local Counterfactual Explanations

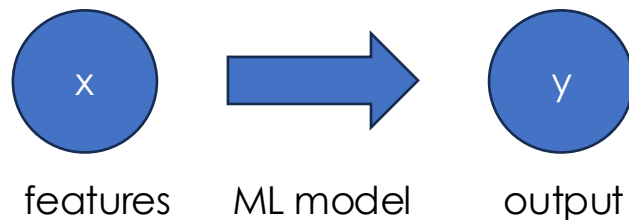
Causality



Find a **cause X** for the **effect Y**

- what does cause-effect mean?
- Necessary condition ($X \Leftarrow Y$)
- Sufficient condition ($X \Rightarrow Y$)
- **Counterfactual reasoning** ($\neg X \Rightarrow \neg Y$)
“if X had not occurred, then Y would not have occurred”

Counterfactual Explanations



ML model = binary classifier
output = positive/negative

A **counterfactual explanation** of a specific **outcome** describes **changes** to **features** that result in getting the **opposite output**

“If the features were x' , then the model would output $-y$ ”

x = **factual**

x' = **counterfactual**;

$x' - x$ = **counterfactual explanation**, i.e., the change necessary to cause the opposite output

Post Hoc Explainability



Bank



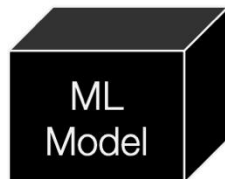
Rosa



Other customers of the bank

Name	Age	Education	Marital	...	Race	Sex	Hours
Rosa	34	Bachelors	Unmarried	...	White	female	40

Rosa's data



83%

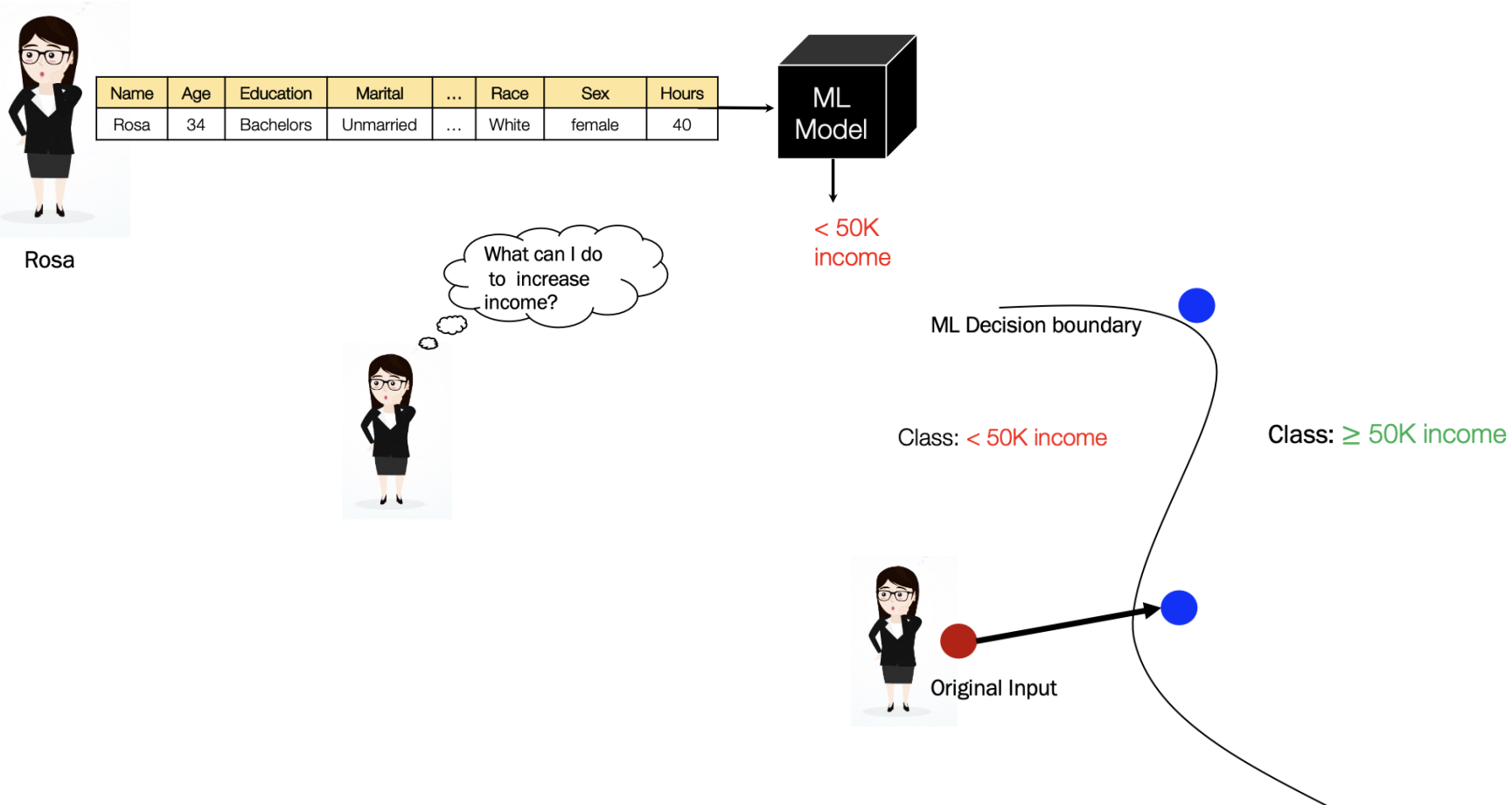
Probability that Rosa's
income < 50K



Manager



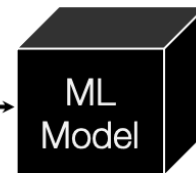
Counterfactual Explanations



Counterfactual Explanations

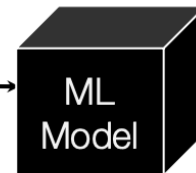


Name	Age	Education	Marital	...	Race	Sex	Hours
Rosa	34	Bachelors	Unmarried	...	White	female	40



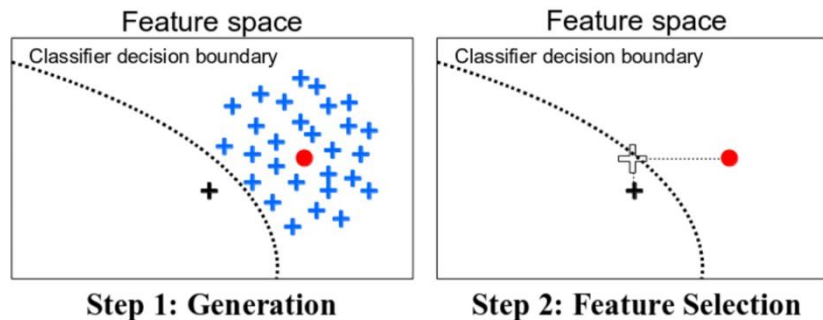
< 50K
income

Name	Age	Education	Marital	...	Race	Sex	Hours
Counterfactual Rosa	34	Bachelors	Unmarried	...	White	female	65

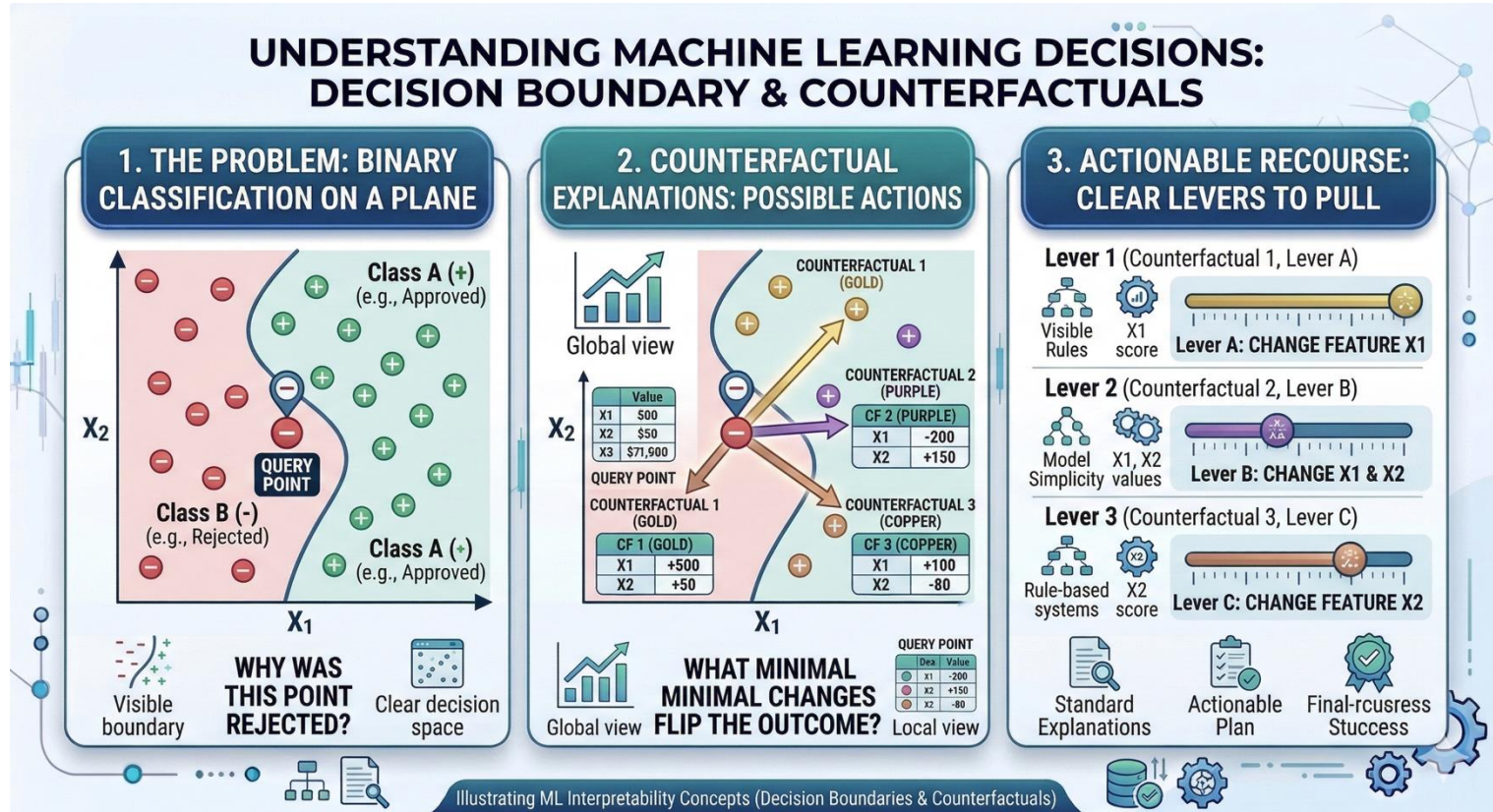


≥ 50K
income

“Inverse Classification”: Identify a close neighbor that has the opposite outcome



Counterfactual Explanations – Decision Boundary

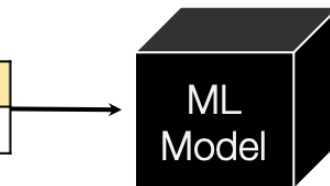


Diverse Counterfactual Explanations

- There are multiple ways to receive the favorable outcome
- Generate multiple diverse counterfactuals



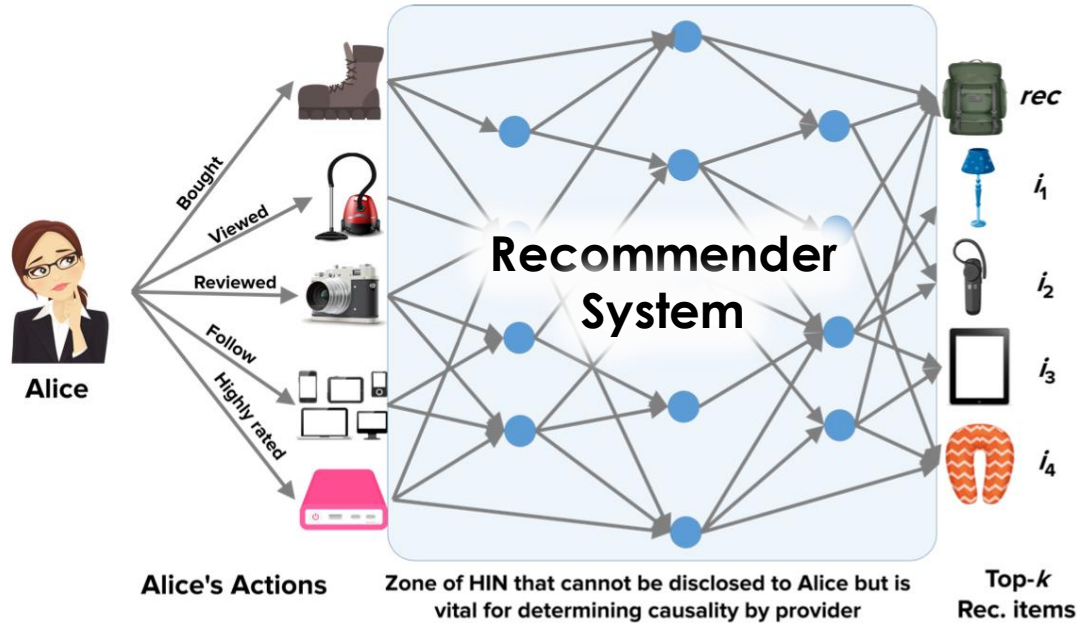
Name	Age	Education	Marital	...	Race	Sex	Hours
Rosa	34	Bachelors	Unmarried	...	White	female	40



Counterfactual Examples by DiCE

1. Increase **hours** to at least 65
2. Change **education** to Masters and increase **work experience** by 1 year

Counterfactual Explanations – Recommendations

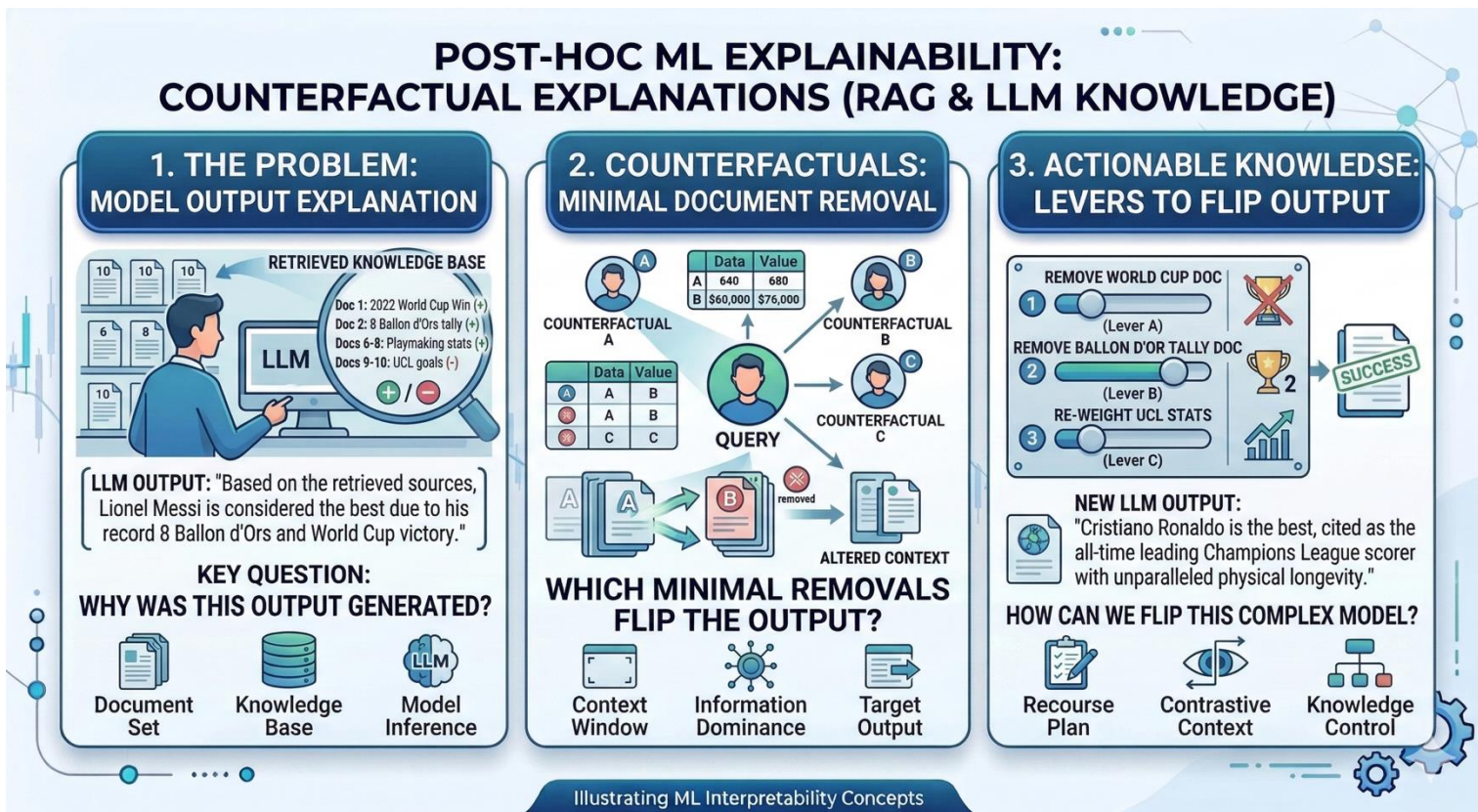


Alice: Why did I receive this recommendation “Jack Wolfskin backpack”?

PRINCE: You **bought** “Adidas Hiking Shoes”;
 You **reviewed** “Nikon Coolpix Camera” with “Sleek! Handy on hikes!”;
 You **rated** “Intenso Travel Power Bank” highly.

If you **had not** done these actions:
 “iPad Air” **would have replaced** “Jack Wolfskin backpack”.

Counterfactual Explanations – LLM+RAG



What is a good Counterfactual Explanation?

A Counterfactual should be as **similar** as possible to the instance regarding feature values

Define an optimization problem

$$L(\mathbf{x}, \mathbf{x}', y', \lambda) = \lambda \cdot (\hat{f}(\mathbf{x}') - y')^2 + d(\mathbf{x}, \mathbf{x}')$$

$\mathbf{x}$: factual

$\mathbf{x}'$: counterfactual

y' : desired outcome

What is a good Counterfactual Explanation?

A Counterfactual Explanation should also be:

- **probable**
 - counterfactual looks like realistic data
- **feasible**
 - changes are reasonable to implement (e.g., do not decrease age)

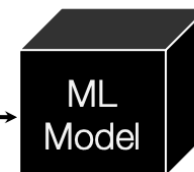
May require a causal model

Algorithmic Recourse

- **Input:** Individual with negative outcome
- **Goal:** Identify smallest cost intervention that can flip the outcome in the future



Name	Age	Education	Marital	...	Race	Sex	Hours
Rosa	34	Bachelors	Unmarried	...	White	female	40



What should I do to flip the outcome?

Algorithmic Recourse

- Provide an **action** for an individual to revert an **unfavorable outcome**
 - e.g., loan rejected.
- **Action** is the change from the factual to a counterfactual:
 - e.g., change education to Master's
- We say that an action **brings recourse** to an individual
- There is a **cost function** that quantifies how **probable** and **feasible** an action is
 - (challenging how to define cost function)

Explainability

Global Counterfactual Explanations

Global Counterfactual Explainability

Global Explainability:

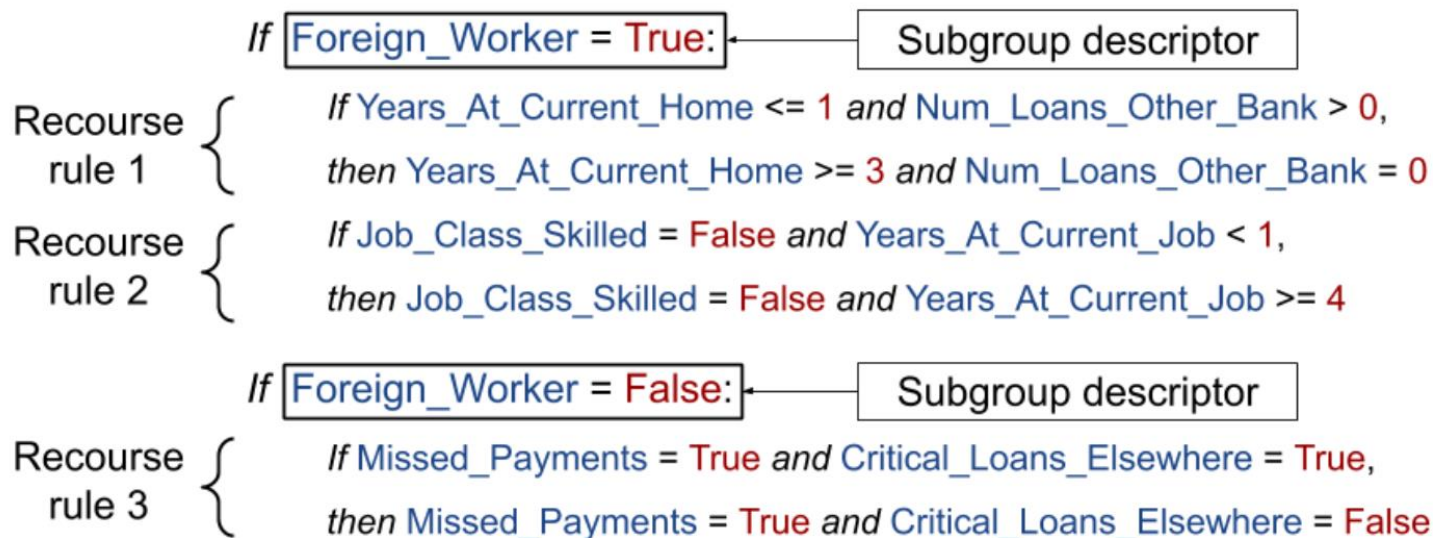
- explain not a specific instance but the **whole model**

Global Counterfactual Explainability: **from local to global**

- **Summarize** counterfactual information for the whole population
- But keep it simple: identify a few counterfactual **actions** that concern a large part of the population

ARES: Actionable Recourses Summary

- Find a small set of **actions** that hold for subpopulations



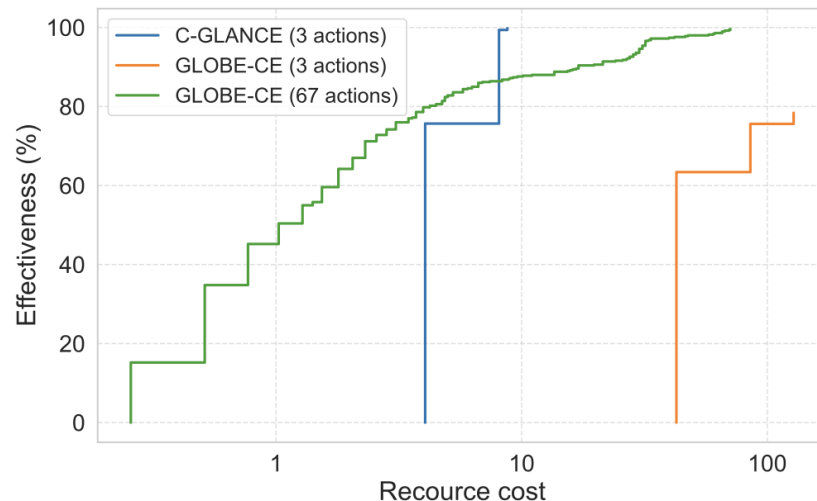
Goal of Global Counterfactual Explainability

In a (sub-)population, an action may not bring recourse to all individuals.

effectiveness is how many achieve recourse via the action

cost of implementing the action

Trade-off between cost-effectiveness



When are global counterfactual explanations (GCEs) good?

Small size: few actions to enhance interpretability

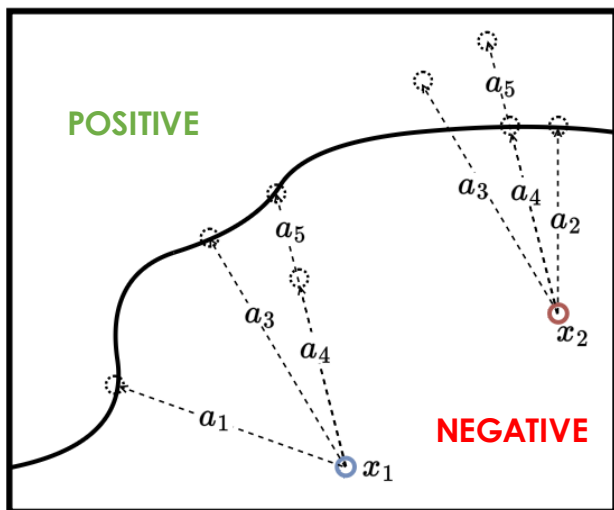
Low cost: actions can be implemented efficiently

High effectiveness: actions cover many individuals

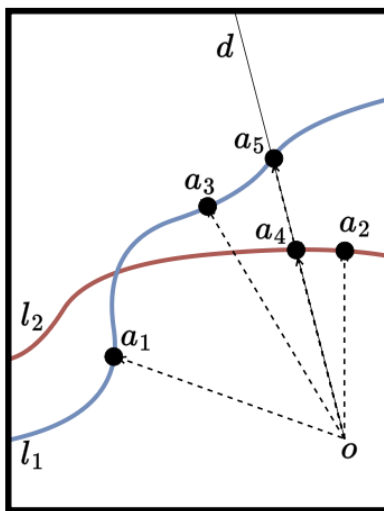
Local vs Global Counterfactual Explainability

individuals x_1, x_2 ; possible actions a_1 - a_5

which action to pick for global explainability?



(a) Feature space



(b) Action space

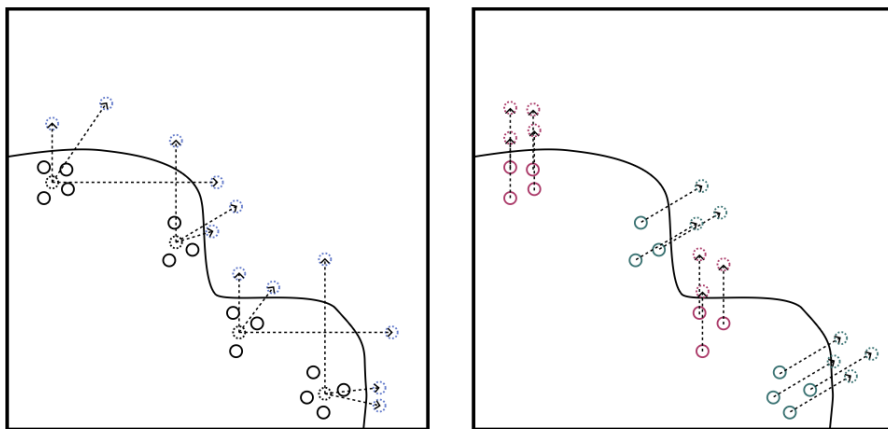
a_1 : best *local* action for x_1

a_2 : best *local* action for x_2

a_3 : best *global* action for x_1 and x_2

GLANCE: Global Actions in a Nutshell for Counterfactual Explainability

A joint clustering on the feature and action space



1. Cluster on feature space
2. Create diverse candidate actions from cluster centroid
3. Re-cluster on feature and action space

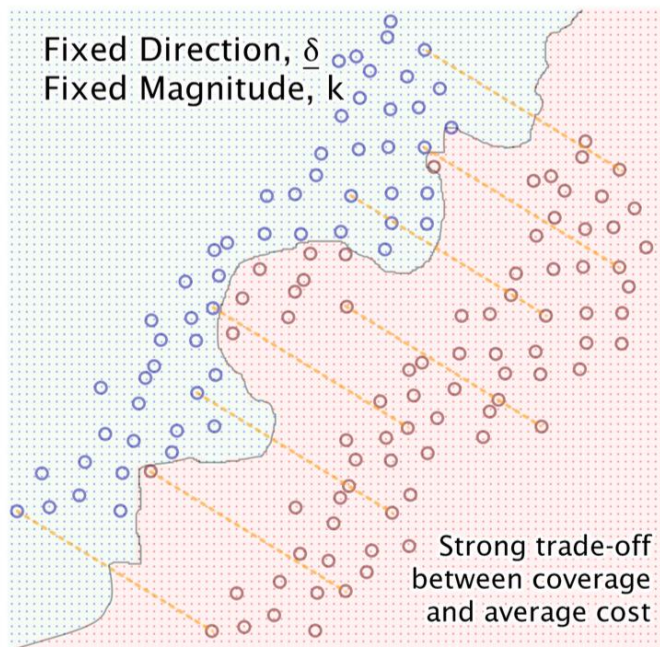
Implicit GCE: each individual to select the most cost-effective action from a global set

Explicit GCE: partition individuals by features, assigning a single action per partition

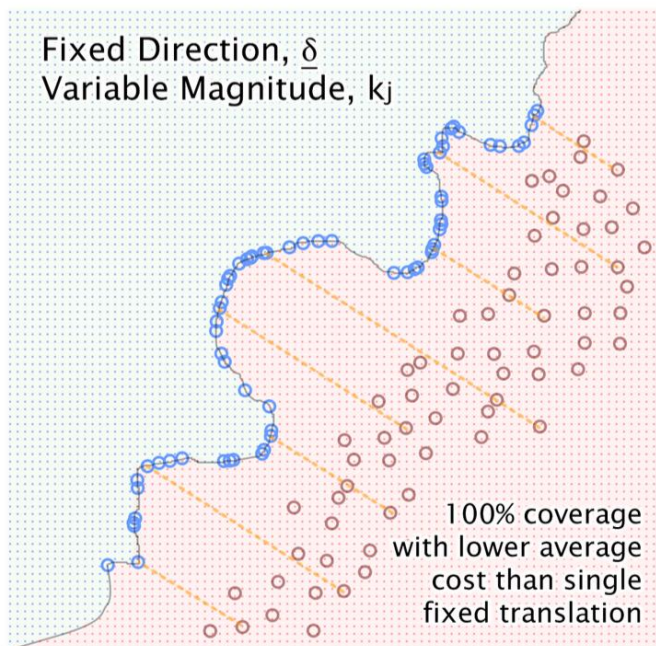
GLOBE-CE: Translation-Based Global Counterfactual Explanations

global **action** vs. global **direction** of local actions

Fixed Translations (Prior Work)



Scaled Translations (Ours)

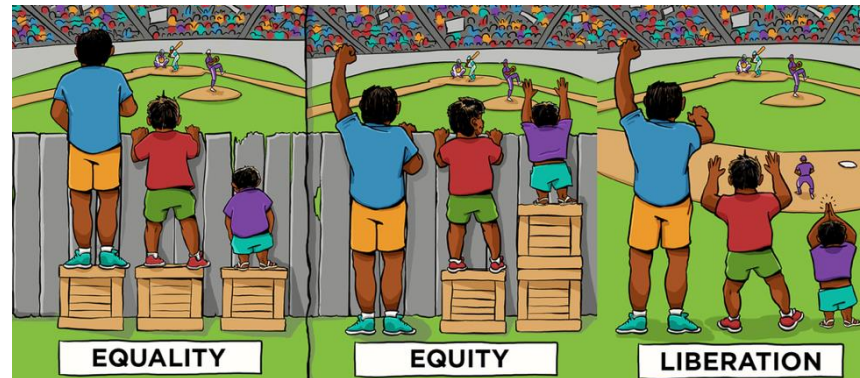


Fairness

Algorithmic Fairness

- AI Systems often make **decisions** on behalf of humans
 - high-stake decisions: likelihood to reoffend; grant a loan application
 - innocuous decisions: which article to read; what to buy in a store
 - or not so innocuous? fake news, filter bubbles

Algorithmic Fairness: machine made decisions should *not discriminate* against individuals



Group vs Individual Fairness

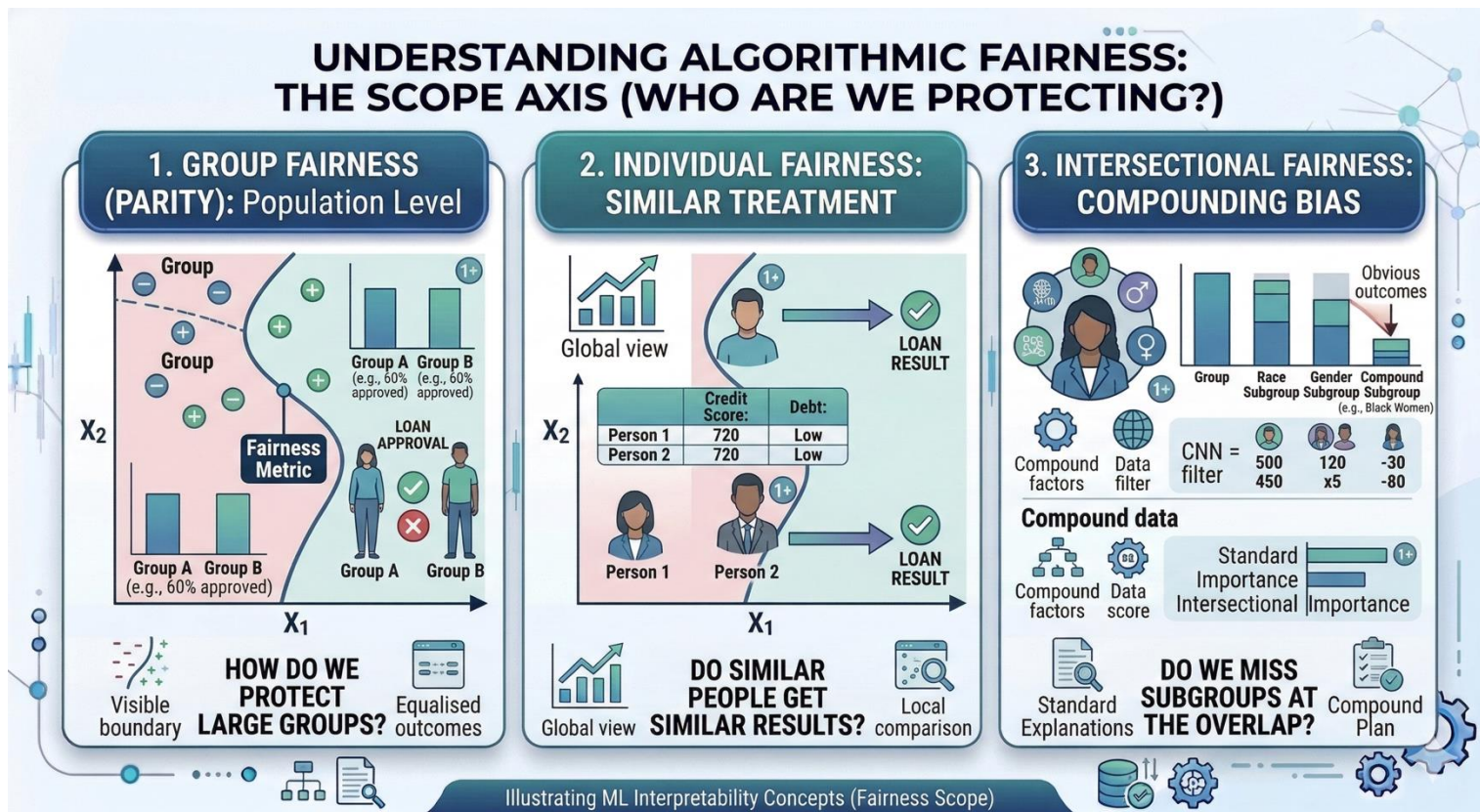
Individual Fairness

- a specific **individual** should be treated/impacted similarly to a **similar other** individual

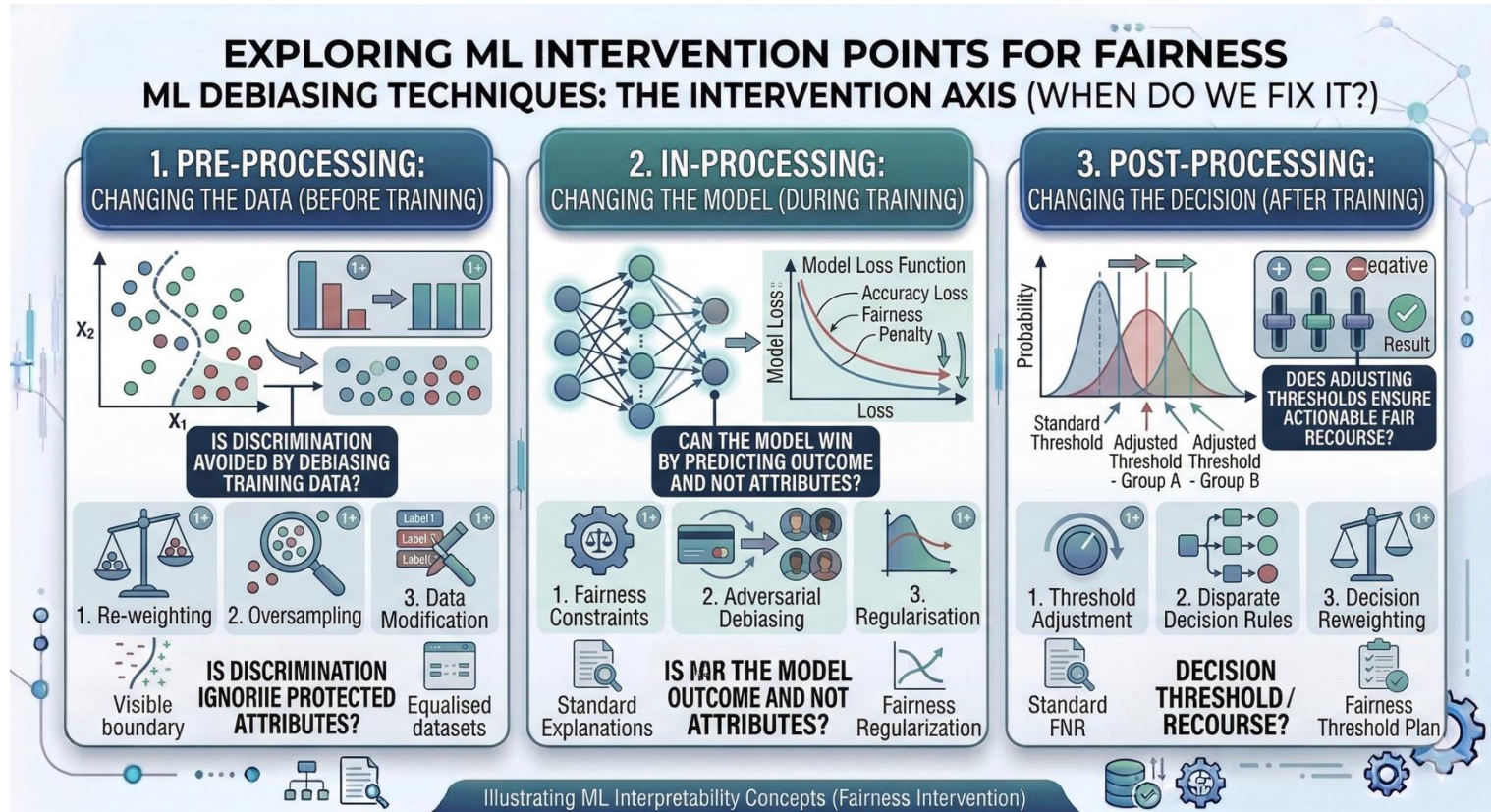
Group Fairness

- a specific **protected** group defined by a **sensitive** attribute (e.g., race, gender) should be treated/impacted similarly to the other (non-protected) groups
 - typically two groups protected (disadvantaged) and non-protected (advantaged)

Group vs Individual Fairness



How to Mitigate Unfairness



Fairness

Individual Level

Counterfactual Fairness

From **individual fairness**:

“similar individuals should get similar predictions”

$$\hat{Y}(X^{(i)}, A^{(i)}) \approx \hat{Y}(X^{(j)}, A^{(j)})$$

To **counterfactual fairness**:

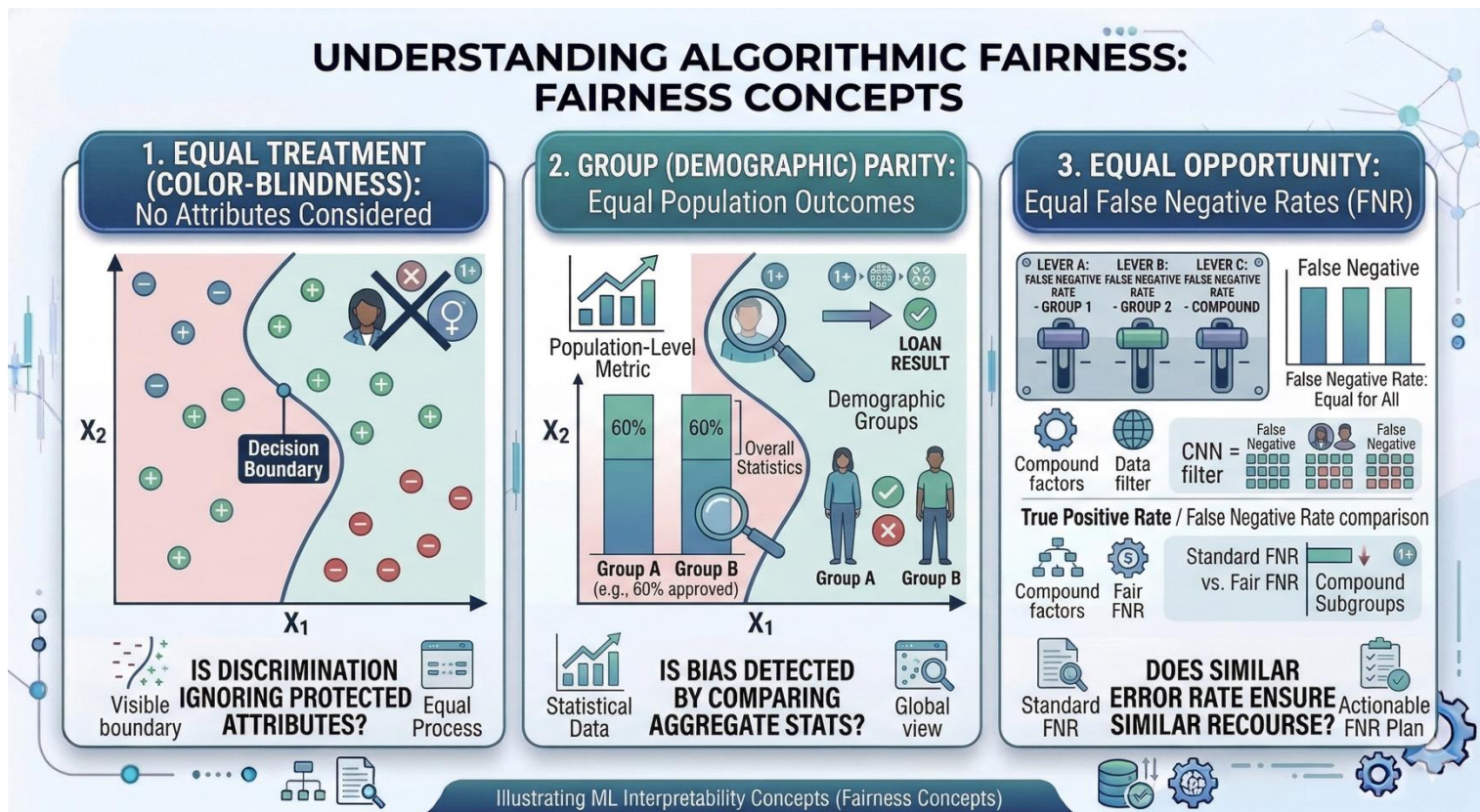
“if my protected value was changed, I should get the same predictions”

$$P(\hat{Y}_{A \leftarrow a}(U) = y \mid X = x, A = a) = P(\hat{Y}_{A \leftarrow a'}(U) = y \mid X = x, A = a)$$

Fairness

Group Level

Group Fairness



Group Fairness Notions

- **Equal treatment** (color-blindness, fairness by unawareness): same threshold
- **Group parity** (demographic or statistical parity): same positive rate
- **Equal opportunity**: same true positive rate (recall)
- You cannot have it all!
- There are various impossibility theorems related to fairness in classification

Causal Group Fairness

- **Group Fairness** requires that predictions are **independent** of protected attribute
 - “no correlation”

But having a causal model allows us to formulate a stronger requirement

- **Causal Group Fairness** requires that the protected attribute **does not directly influence** the predictions
 - “no causation”

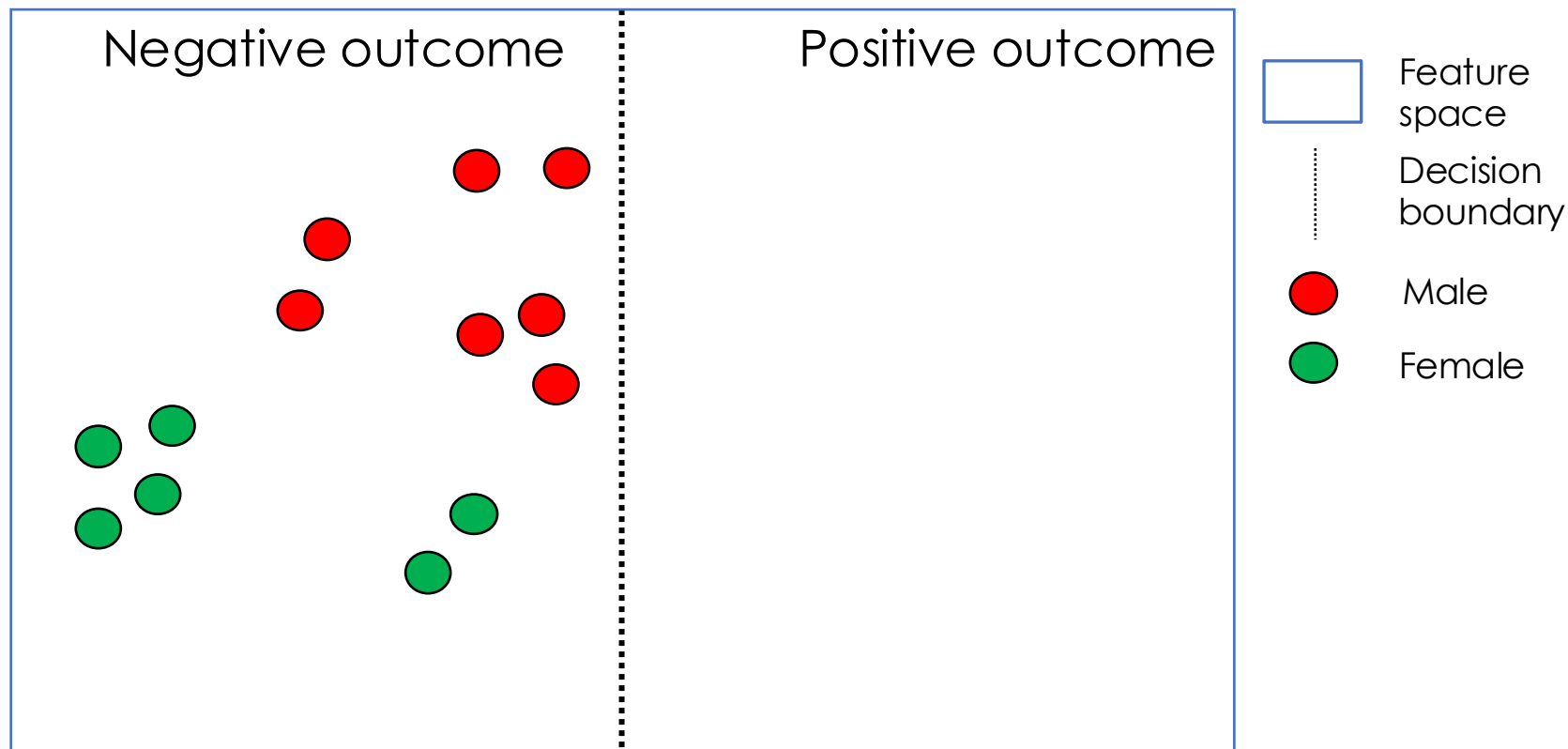
Discover Unfairness via Counterfactuals

- Explanations can uncover cases of unfairness
- A subpopulation x **does not receive** a loan
 - $x = \{\text{Race}=\text{Caucasian}, \text{Sex}=\text{Female}, \text{Income}=\text{Low}\}$
- What are the **minimal changes** to **receive** the loan?
 - $\{\text{Sex}=\text{Female}\} \rightarrow \{\text{Sex}=\text{Male}\}$
- This shows **discrimination** of x based on sex!

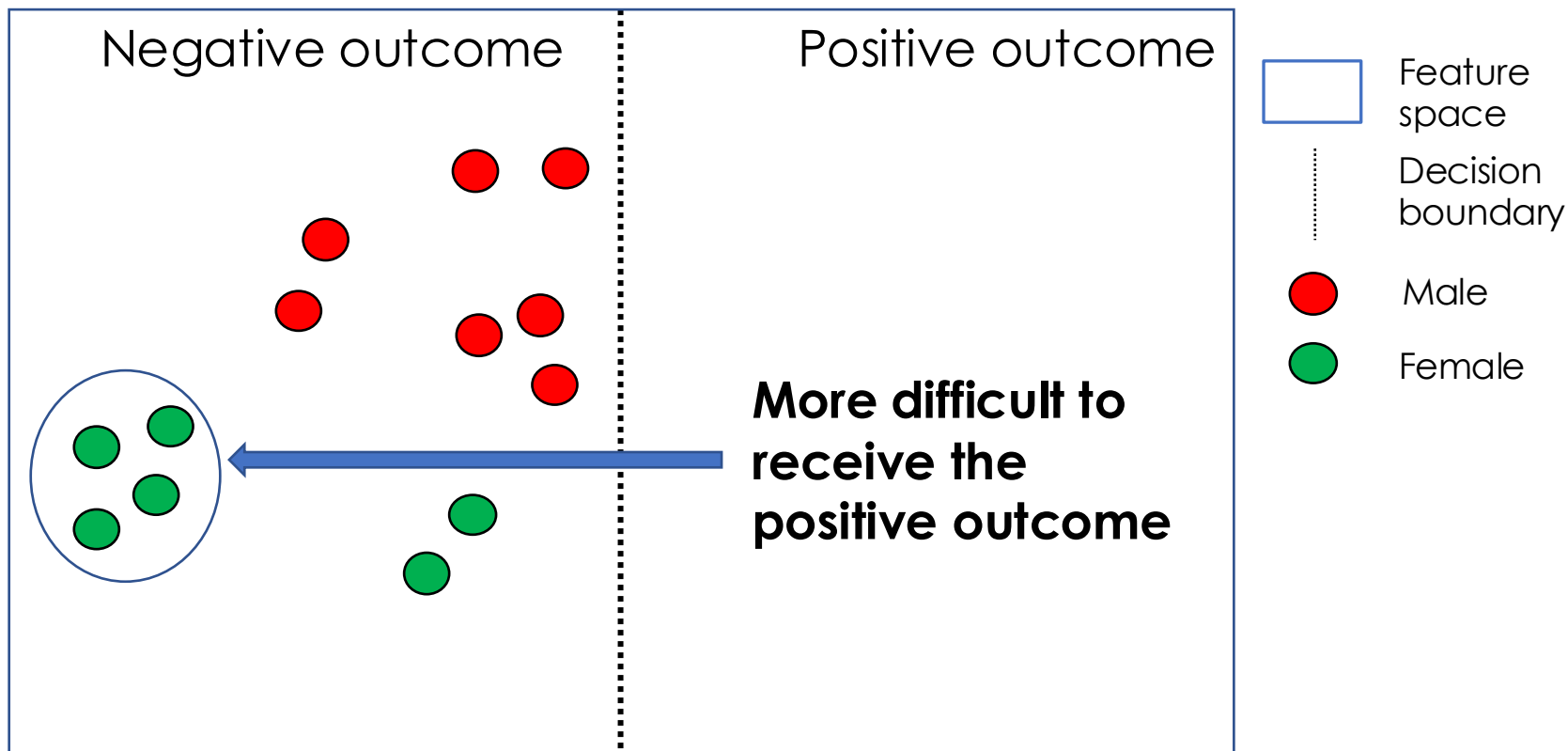
Fairness of Recourse

- **recourse** = the way to reverse an unfavorable outcome
- **fairness of recourse** = does the recourse cost the same?

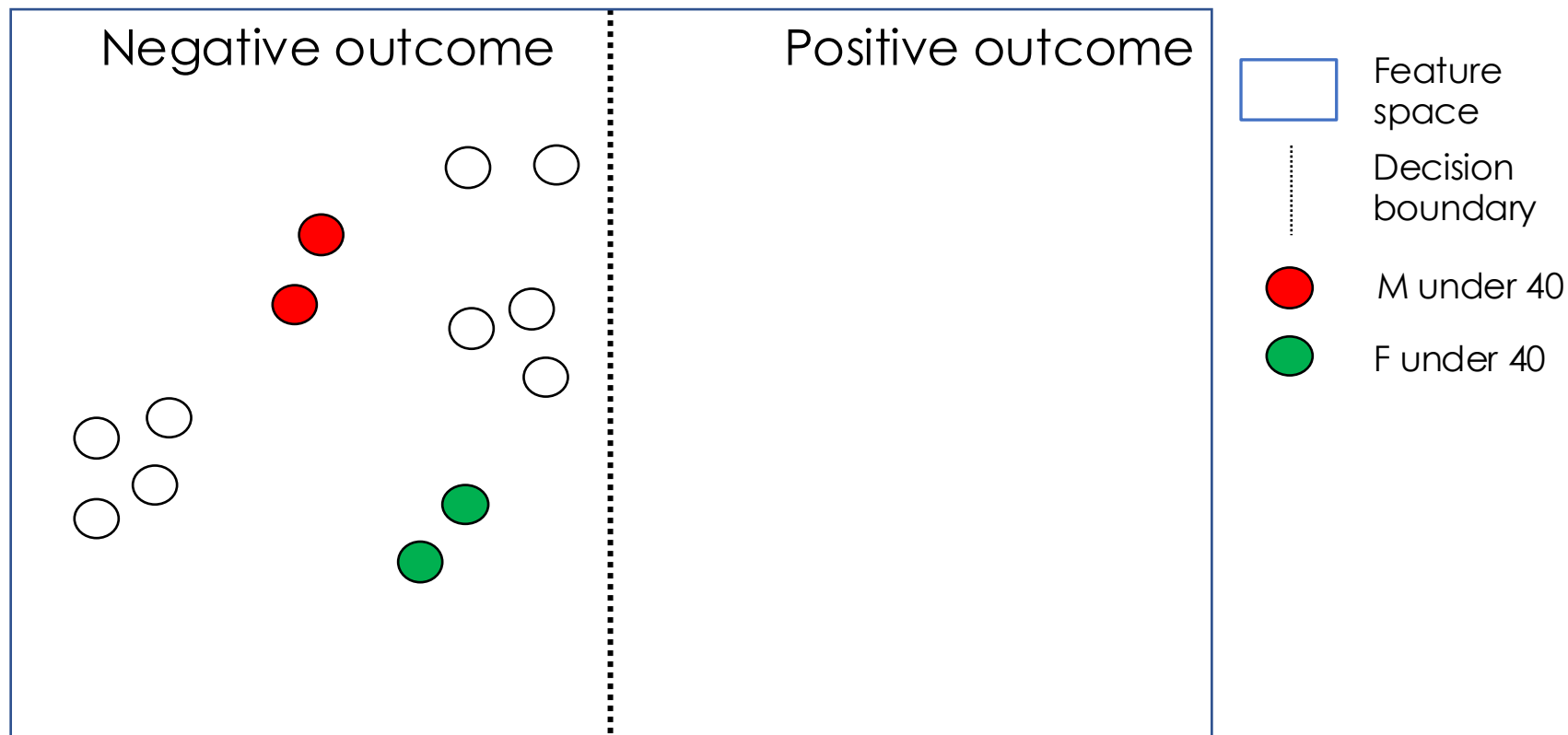
FACTS: Fairness-Aware Counterfactuals for Subgroups



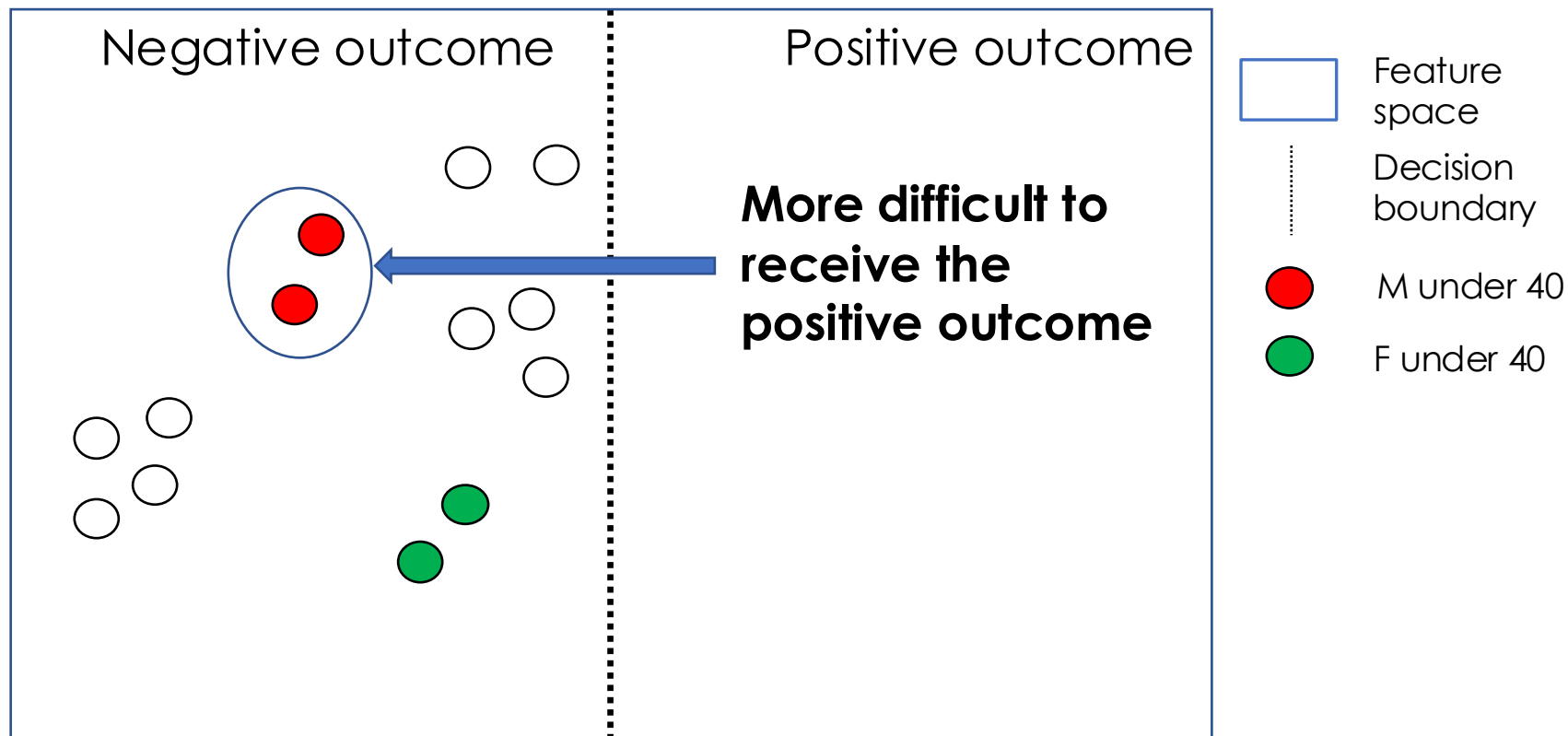
FACTS: Fairness-Aware Counterfactuals for Subgroups



FACTS: Fairness-Aware Counterfactuals for Subgroups



FACTS: Fairness-Aware Counterfactuals for Subgroups



FACTS: Fairness-Aware Counterfactuals for Subgroups

- Introduced various notions of recourse fairness that compare the **cost-effectiveness distributions** of the protected subgroups
- **Burden** [prior def]: “The mean recourse cost is the same”
- **Equal effectiveness**: “The same proportion of individuals in the protected subgroups can achieve recourse”
- **Equal Effectiveness within Budget**: “The same proportion of individuals in the protected subgroups can achieve recourse with a cost at most x ”
- **Equal Cost of Effectiveness**: “The minimum cost to achieve $x\%$ effectiveness is the same”
- **Equal Choice for Recourse**: “Protected subgroups have the same number of options to achieve recourse with $x\%$ effectiveness”

FACTS: Fairness-Aware Counterfactuals for Subgroups

If **capital-gain** = 0, **education-num** = 10, **hours-per-week** = FullTime, **race** = White:

Protected Subgroup 'Male', 8.40% covered

Make **hours-per-week** = OverTime with effectiveness 8.92% and counterfactual cost = 2.0.

Make **education-num** = 11 with effectiveness 8.92% and counterfactual cost = 3.0.

Make **hours-per-week** = BrainDrain with effectiveness 16.98% and counterfactual cost = 4.0.

Make **education-num** = 11, **hours-per-week** = OverTime with effectiveness 23.16% and counterfactual cost = 5.0.

Make **education-num** = 12 with effectiveness 23.16% and counterfactual cost = 6.0.

Make **education-num** = 12, **hours-per-week** = OverTime with effectiveness 31.56% and counterfactual cost = 8.0.

Aggregate cost of the above recourses = 8.00

Protected Subgroup 'Female', 8.32% covered

Make **hours-per-week** = OverTime with effectiveness 1.79% and counterfactual cost = 2.0.

Make **education-num** = 11 with effectiveness 1.79% and counterfactual cost = 3.0.

Make **hours-per-week** = BrainDrain with effectiveness 3.27% and counterfactual cost = 4.0.

Make **education-num** = 11, **hours-per-week** = OverTime with effectiveness 4.46% and counterfactual cost = 5.0.

Make **education-num** = 12 with effectiveness 4.46% and counterfactual cost = 6.0.

Make **education-num** = 12, **hours-per-week** = OverTime with effectiveness 5.95% and counterfactual cost = 8.0.

Aggregate cost of the above recourses = inf

Bias against Female. Unfairness score = inf.

Thank you!

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